

Rahul Ray

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MSc Data Science student at Leiden with a habit of building things to answer questions I couldn't find clean answers to.
Interested in what actually happens when you push a method past its assumptions.

Education

MSc Computer Science – Data Science *Leiden University, Netherlands* Sep 2025 – Aug 2027 (expected)
BTech Computer Science *SRM Institute of Science and Technology, Chennai* 2021 – 2025

Work Experience

Machine Learning Intern *Mar – Aug 2025 · Fox Solutions Pvt. Ltd., India*

- Developed a ViT-based computer vision pipeline end-to-end, from model training through to edge deployment with ONNX, optimising for inference latency and memory footprint on constrained hardware.

Projects

Can LLMs Make Tabular ML Better?

[writeup](#) [GitHub](#)

Python · XGBoost · LLaMA 3.1 · sentence-transformers · Optuna · SHAP

- Tested LLM-generated semantic features as input to XGBoost via nested CV (3×5-fold, leak-free PCA, Optuna); the hypothesis failed on both datasets — documented honestly with Wilcoxon tests, Cohen's d, and SHAP analysis.

AutoInsight – End-to-End AutoML Web App

[writeup](#) [GitHub](#)

Python · Streamlit · XGBoost · LightGBM · CatBoost · SHAP · Optuna · Groq API

- Trains 7 models in parallel with stratified CV, SHAP explainability, and Bayesian tuning; privacy-preserving LLM integration (aggregated metrics only, never raw data); live on Hugging Face Spaces.

Quake – Learned Heuristic A* on BSP Geometry

[writeup](#) [GitHub](#)

Python · PyTorch · XGBoost · NetworkX · BSP v29 parsing

- Parsed Quake 1 BSP maps from scratch, generated 625k Dijkstra ground-truth pairs, and trained MLP/XGBoost heuristics cutting A* node expansions by 42% with paths within 2% of optimal.

Epoch-wise Double Descent Under Label Noise – ResNet-18 / CIFAR-10

[writeup](#) [GitHub](#)

Python · PyTorch · ResNet-18 · CIFAR-10

- Reproduced Nakkiran et al. (2019) under η up to 30%, isolating the corrupted-accuracy inflection as the mechanistic peak marker; label smoothing hurt, Mixup partially helped — both reported honestly.

Newsphere Live – Semantic 3D News Map

[writeup](#) [GitHub](#)

Python · sentence-transformers · UMAP · HDBSCAN · Three.js · GitHub Actions

- 11 RSS feeds fetched every 6 hours, embedded via MiniLM, reduced with UMAP, clustered with HDBSCAN, rendered in Three.js — fully serverless, zero backend, zero cost, via GitHub Actions.

JEPA-IR – Benchmarking I-JEPA for Image Retrieval

[writeup](#) [GitHub](#)

Python · PyTorch · I-JEPA ViT-H/14 · CLIP · FAISS

- Benchmarked I-JEPA vs CLIP vs ResNet-50 as frozen extractors on Oxford5k/Paris6k; I-JEPA trailed by ~30 mAP points and near-zero query expansion gain revealed a fundamental embedding space issue, not a tuning problem.

Technical Skills

Core	Python, PyTorch, pandas, NumPy, scikit-learn, experiment design, statistical testing, nested CV
Tabular ML	XGBoost, LightGBM, CatBoost, SHAP, Optuna, feature engineering & selection, Wilcoxon, Cohen's d
Deep Learning	Vision Transformers, ResNet/CNNs, self-supervised learning, Mixup, label smoothing, generalisation theory
RL	Q-Learning, DQN, Policy Gradients, PPO, Actor-Critic, Gymnasium
NLP	sentence-transformers, TF-IDF, BM25, c-TF-IDF, topic modelling, NER, UMAP, HDBSCAN
IR	FAISS, dense retrieval, CLIP, multimodal retrieval, mAP, NDCG, Recall@k, query expansion, re-ranking
MLOps	Automated ML pipelines, edge deployment, model compression, heuristic & graph search, Hugging Face, Streamlit

Languages

English – C1 Professional Proficiency (*IELTS Certified*) | **Hindi** – Native